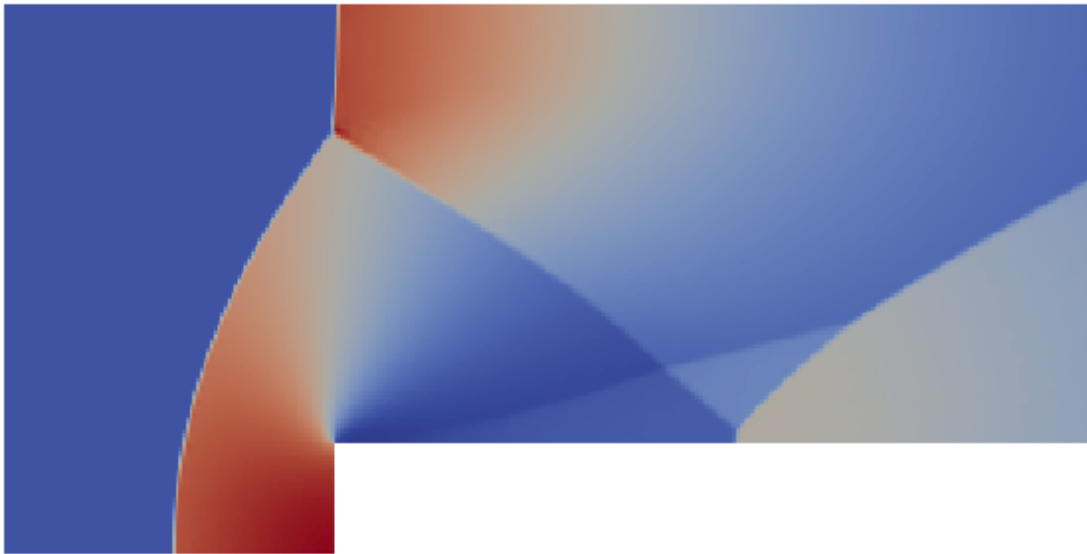


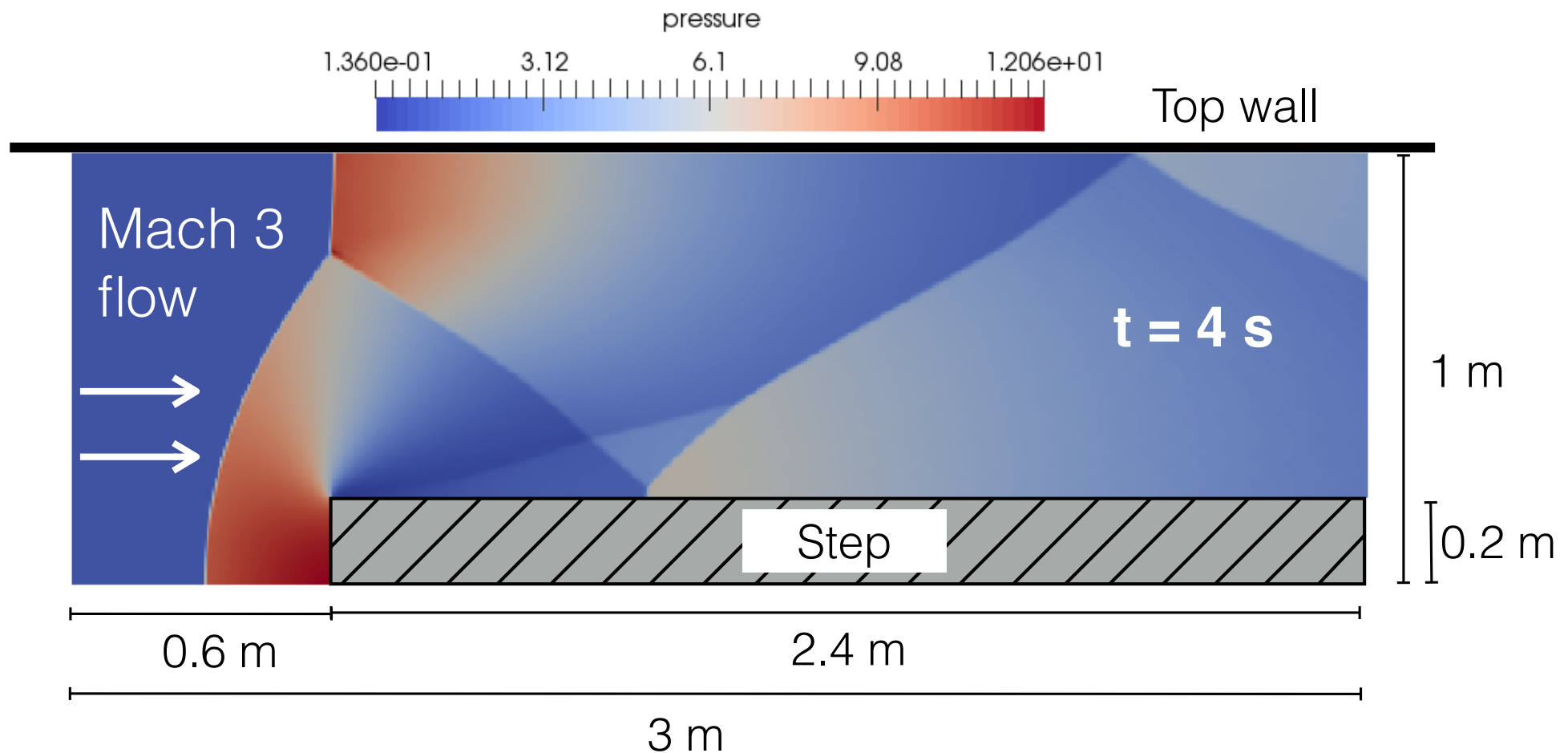
The “Mach 3 wind tunnel with a step”



- A canonical, **two-dimensional**, compressible flow
- In principle, fully characterized by the geometry and the approach Mach number.

$$\text{Ma} = U/a$$

Overview of the flow



Inlet conditions

$$p = 1$$

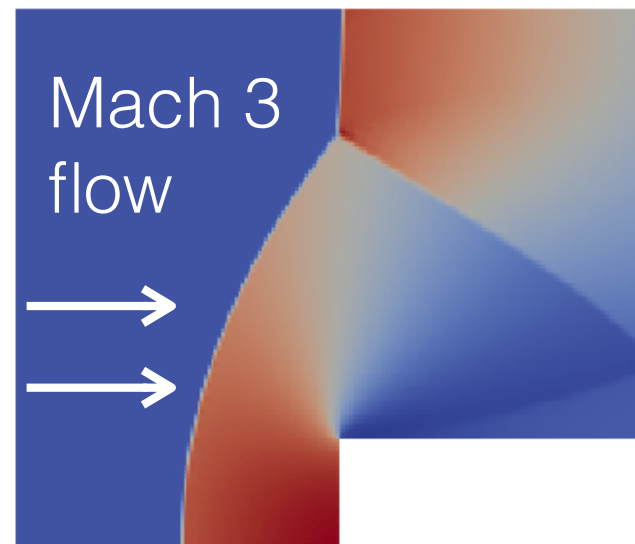
$$\rho = \gamma = 1.4$$

$$(u, v) = (U, 0)$$

$$a = (\gamma p / \rho)^{1/2} = 1$$

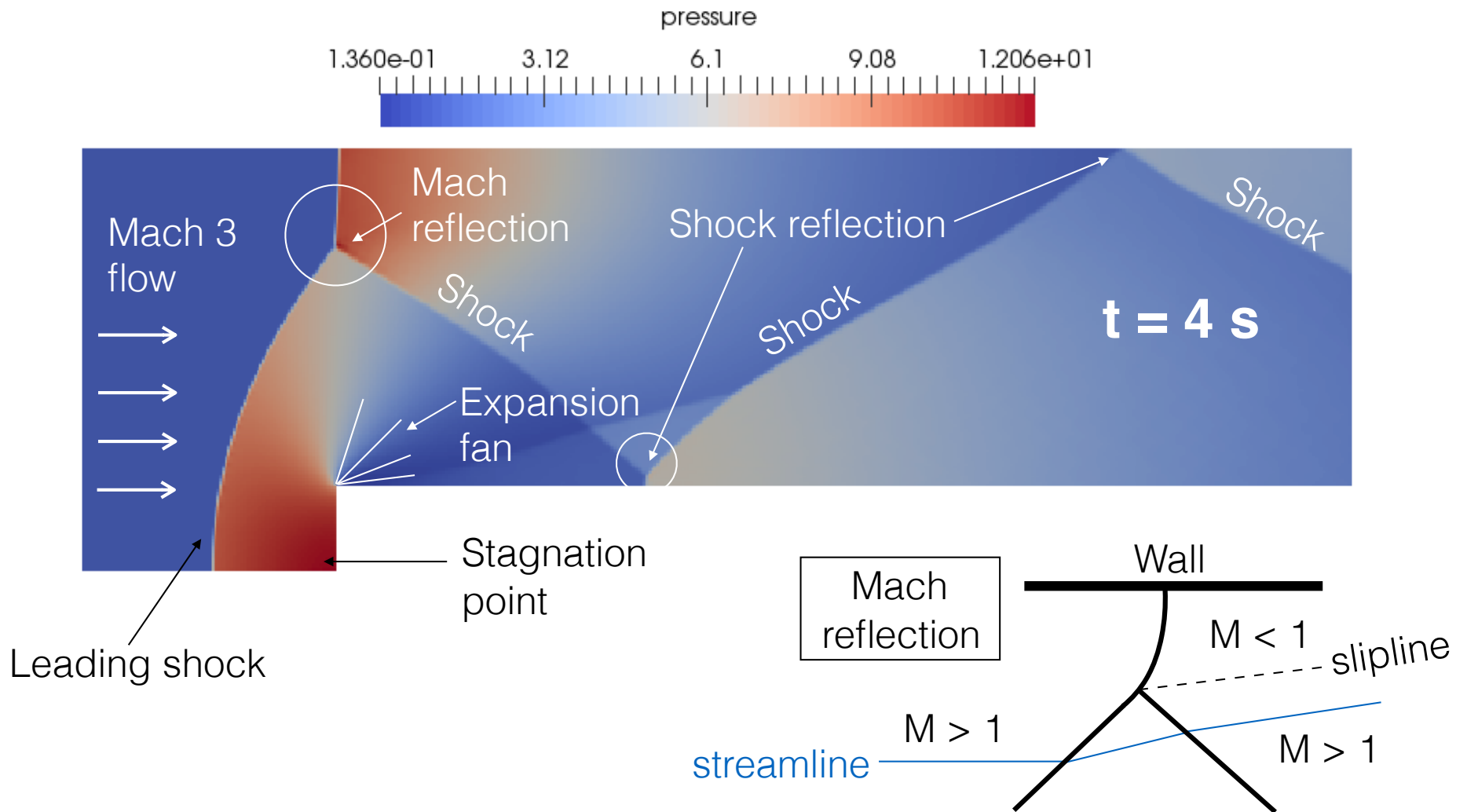
$$M = U/a = 3$$

$$U = 3$$

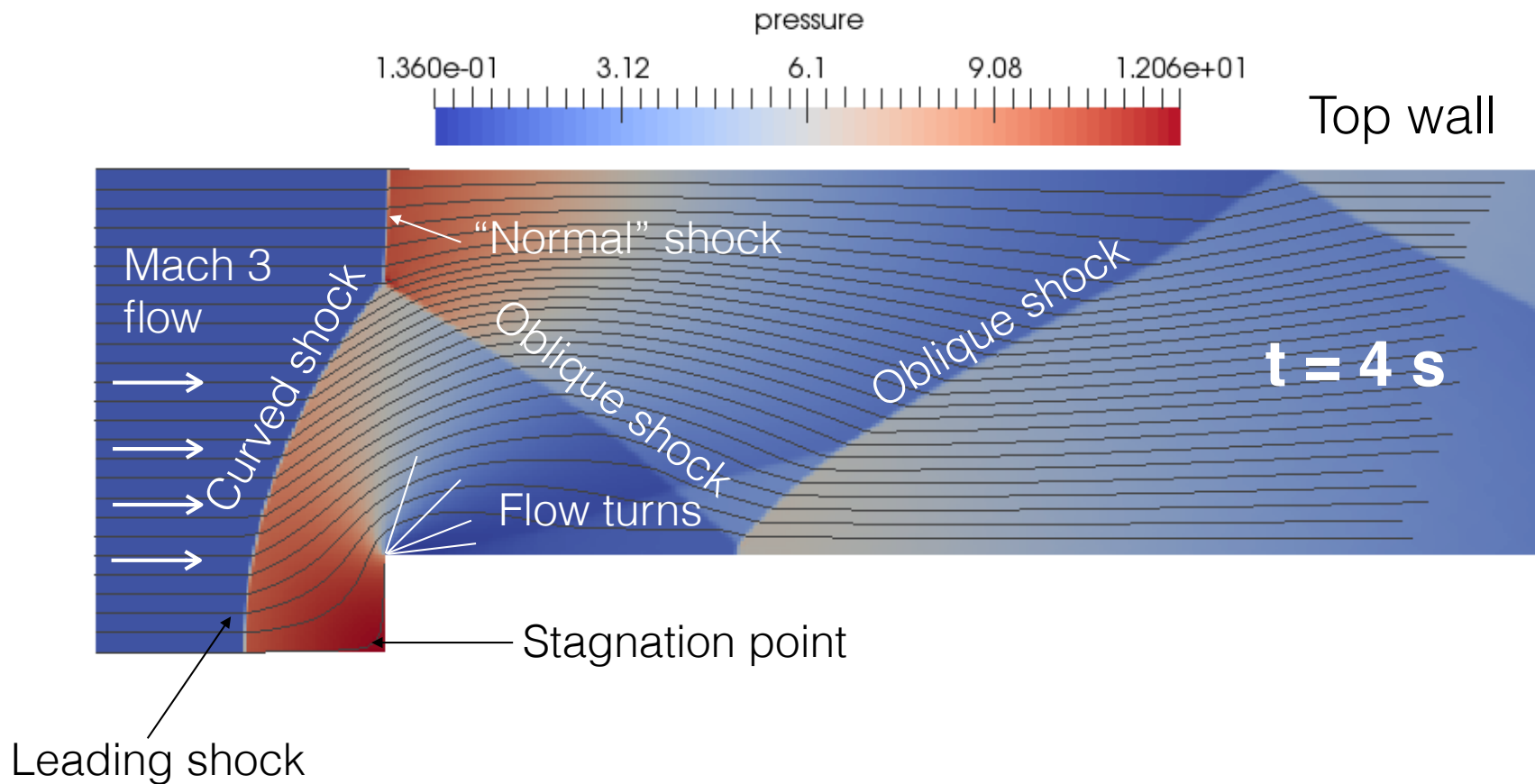


All in base SI units: m, s, Pa, K, kg/m³, etc.

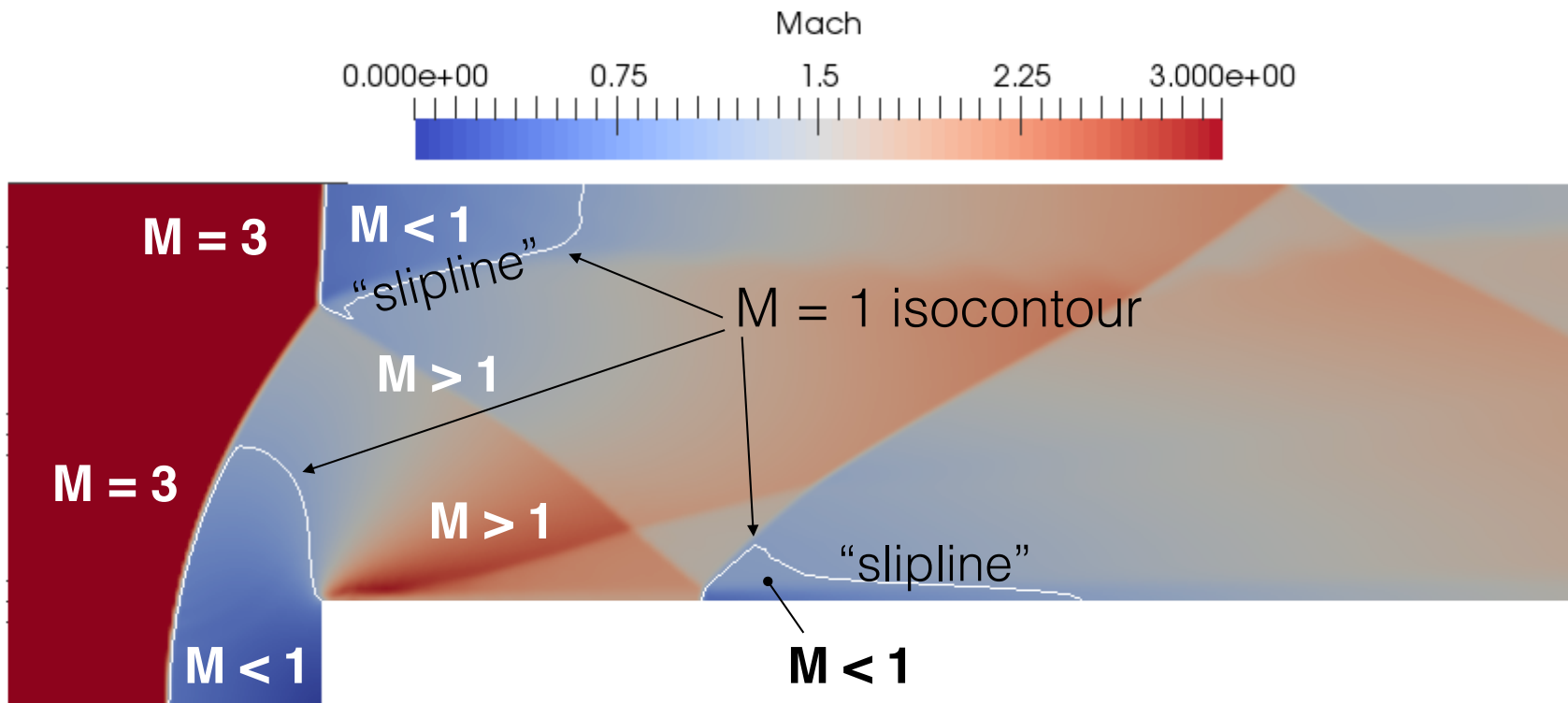
Some details



Flow streamlines



Mach nr

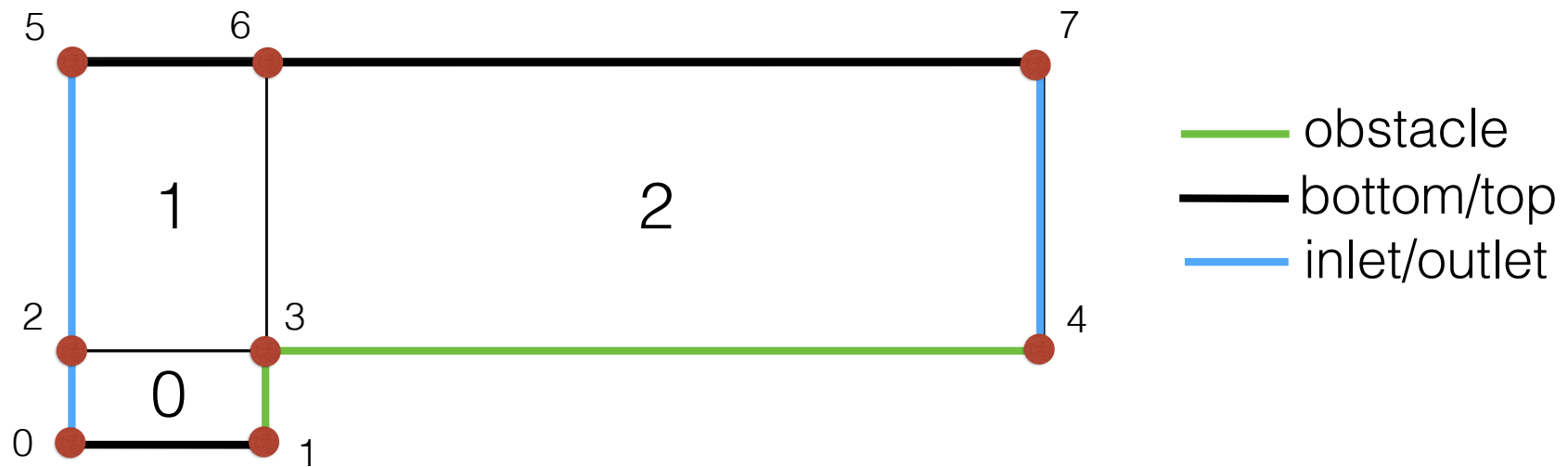


$$M = |\mathbf{u}|/a = |\mathbf{u}|/\sqrt{\gamma p/\rho}$$

Movie

- See Movie online

Meshing strategies



- 16 points (8 x 2) and 3 blocks are sufficient
- Uniform meshes are required, since shocks are “everywhere”
- Make sure to identify faces for boundary conditions

Solver

- You will be using `rhoCentralFoam`
- Solver implements “fast” high-order difference methods for hyperbolic conservation laws, i.e. the **Euler equations**

Field variables

- Velocity vector field, $U = (u,v)$
- Temperature scalar field, T
- Pressure scalar field, p

“Normalized fluid”

- An inviscid fluid with “desirable properties”
- Ratio of specific heats: $\gamma = 7/5 = 1.4$
- Temperature is 1 K for sound speed 1 m/s:

$$a = \sqrt{\gamma p / \rho} = \sqrt{\gamma R T} = \sqrt{\gamma R} = 1 \rightarrow \gamma R = 1$$

$$\gamma \times 8314 \text{ (J/kmol} - \text{K))} / W \text{ (kg/kmol)} = 1$$

$$\rightarrow W = 8314 \times \gamma = 11640 \text{ g/mol}$$

$$\rightarrow c_p = R / (1 - 1/\gamma) = 2.5 \text{ (J/kg} - \text{K)}$$

Inlet conditions

- Exercise caution due to the “normalized fluid” and make sure all conditions are consistent to provide a Mach 3 inlet flow
- $U = (3, 0)$ m/s
 $T = 1$ K
 $p = 1$ Pa
- Since the sound speed at the inlet is $a = 1$ m/s, this is a Mach = 3 flow.
- Also notice that density at the inlet is $\rho = \gamma p / a^2 = \gamma$

Discretization schemes

- More information is available in Kurganov & Tadmor, JCP 2000